

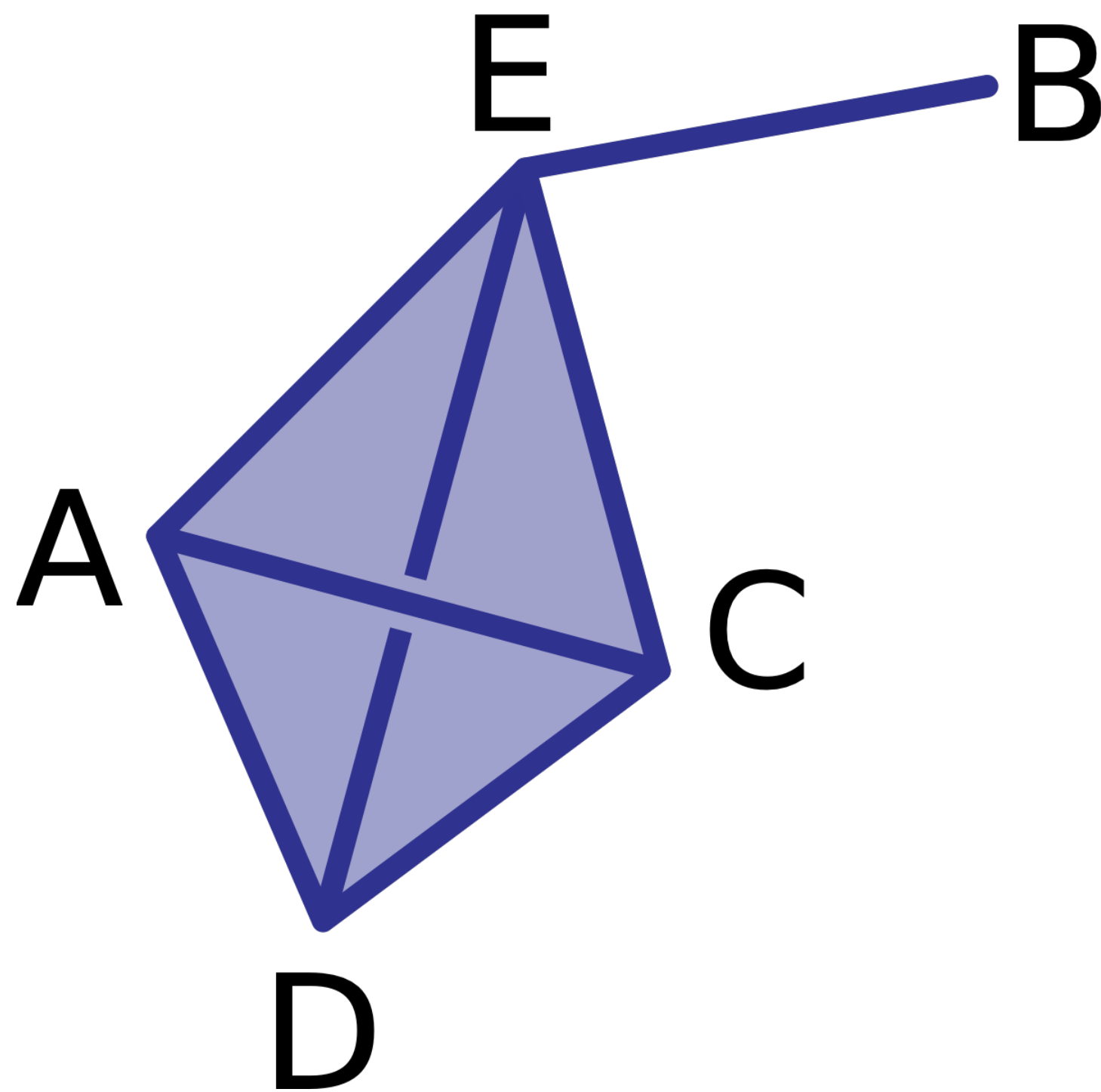
Here Comes the Homology

Will we get there? — Yes!

p -Chains

Let K be a simplicial complex and fix a dimension p .

A p -chain is a finite formal sum of p -simplices in K , written $\alpha = \sum_i a_i \sigma_i$



Chain Group

The collection of p -chains with addition is called the p th-chain group (vector space), $C_p(K)$.

Closure: $c + d \in C_p(K)$

Associative: $(c + d) + e = c + (d + e)$

Commutative: $c + d = d + c$

Zero element: $c + 0 = 0 + c = c$

Inverses: $c + (-c) = 0$

Boundary Maps

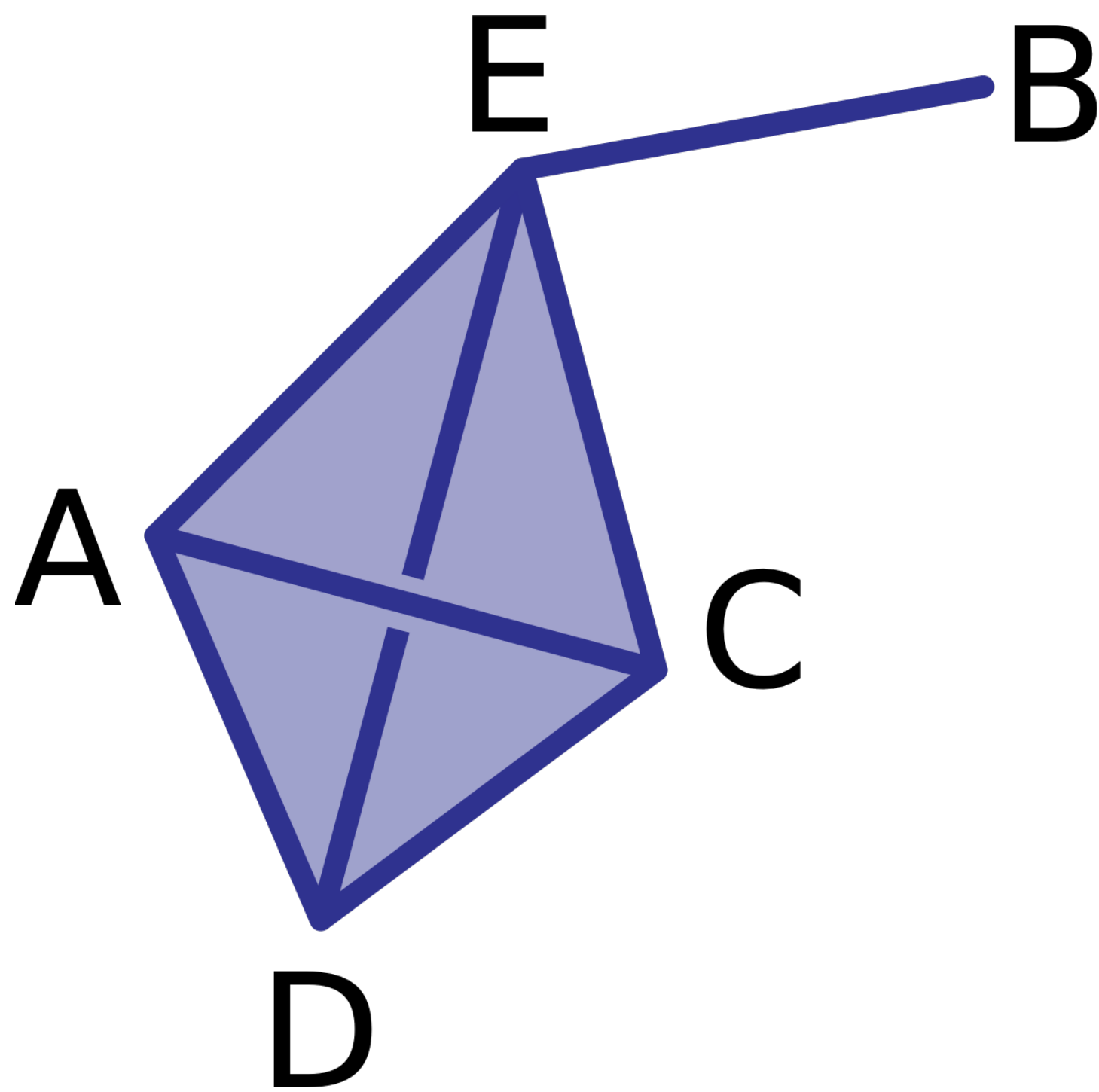
$$\partial_p : C_p(K) \longrightarrow C_{p-1}(K)$$

$$\sigma = [v_0, \dots, v_p] \longmapsto \sum_{j=0}^p [v_0, \dots, \widehat{v_j}, \dots, v_p]$$

Cycles

$$C_{p+1}(K) \xrightarrow{\partial_{p+1}} C_p(K) \xrightarrow{\partial_p} C_{p-1}(K)$$

A chain in the kernel of ∂_p is called a p -cycle. $Z_p(K) = \ker(\partial_p) = \{c \in C_p(K) \mid \partial_p c = 0\}$



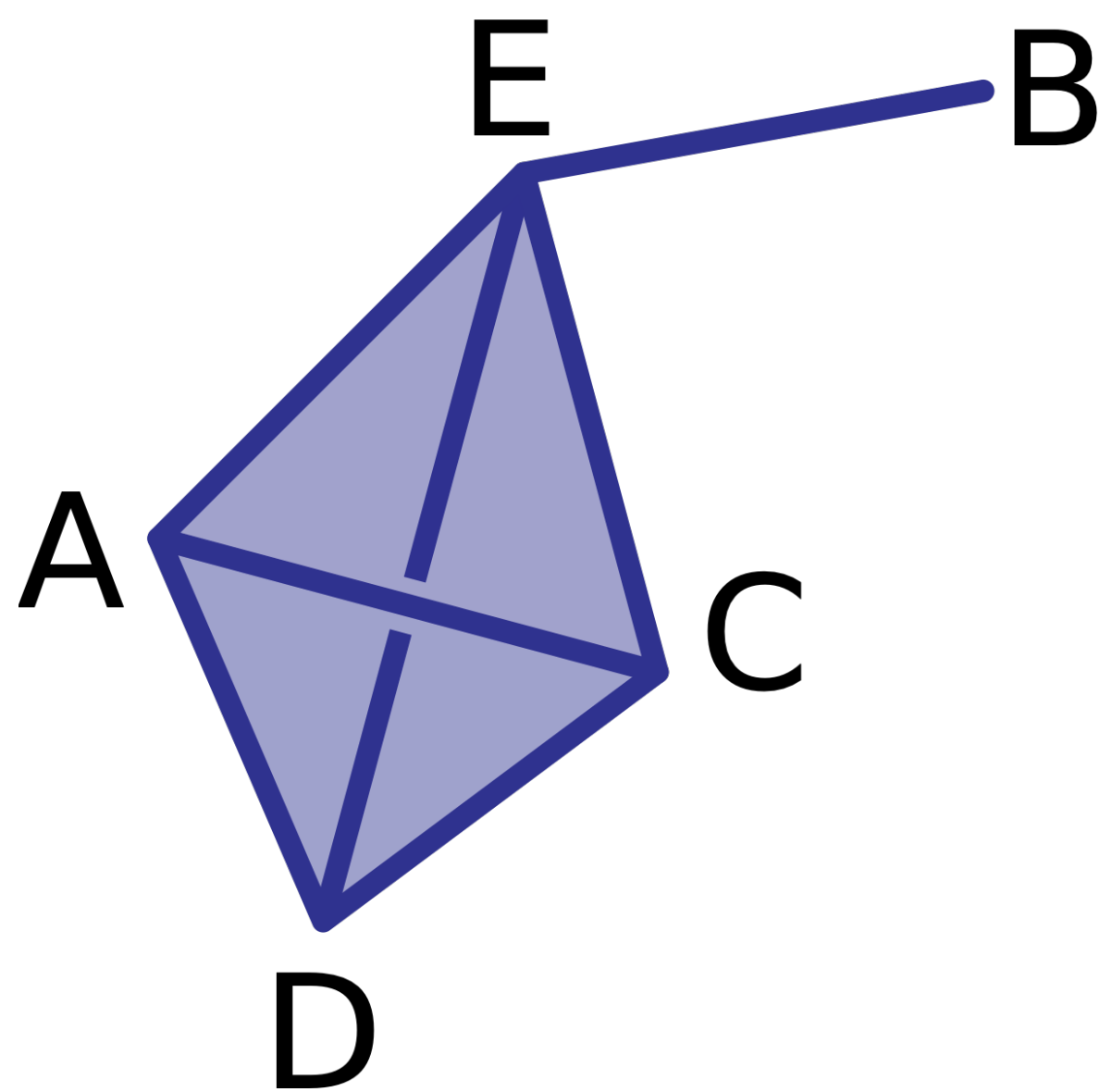
The collection of p -cycles forms a subspace $Z_p(K) \subseteq C_p(K)$

Boundaries

$$C_{p+1}(K) \xrightarrow{\partial_{p+1}} C_p(K) \xrightarrow{\partial_p} C_{p-1}(K)$$

A chain in the image of ∂_{p+1} is called a p -boundary

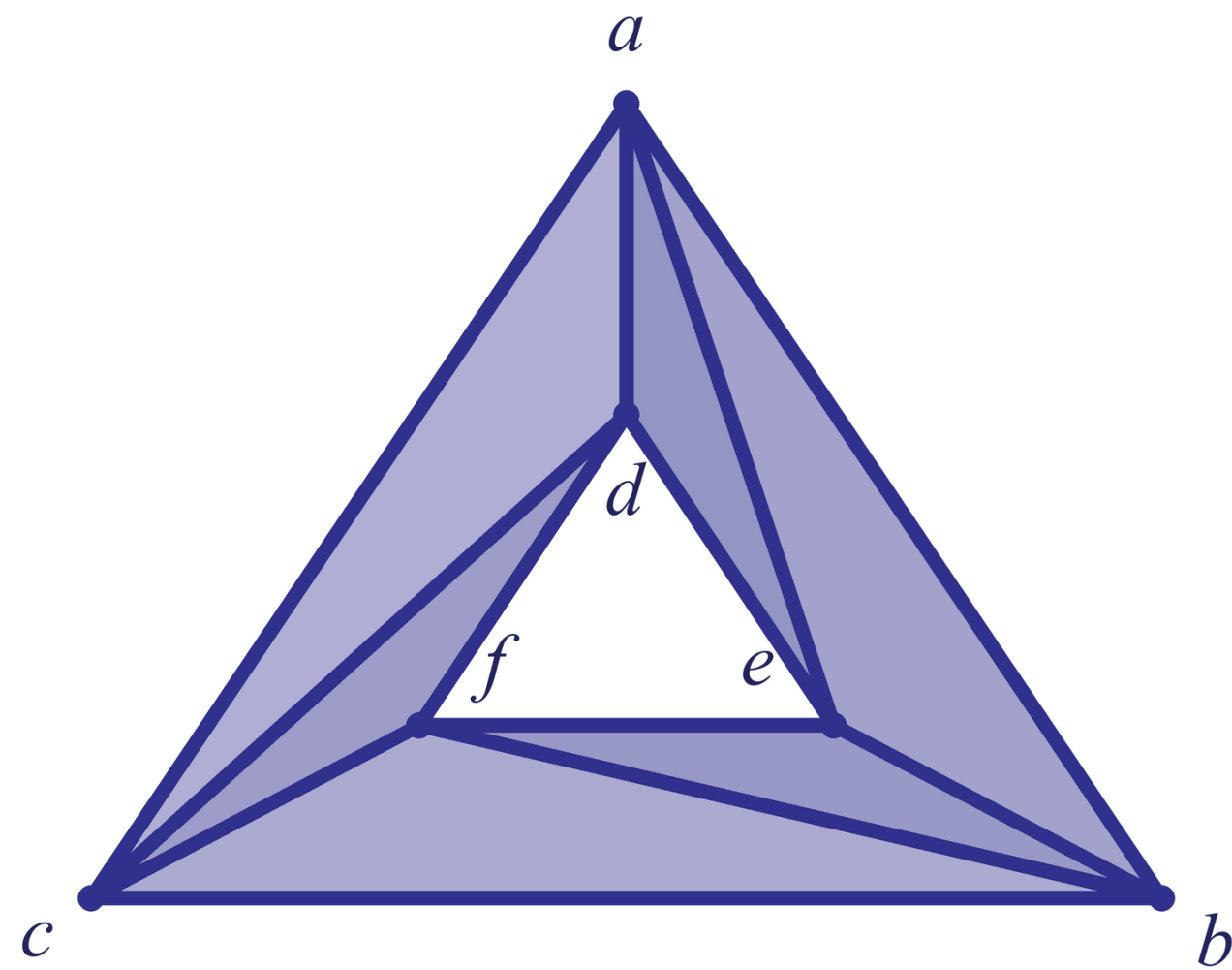
$$B_p(K) = \text{im}(\partial_{p+1}) = \{\partial_{p+1}c \mid c \in C_{p+1}(K)\}$$



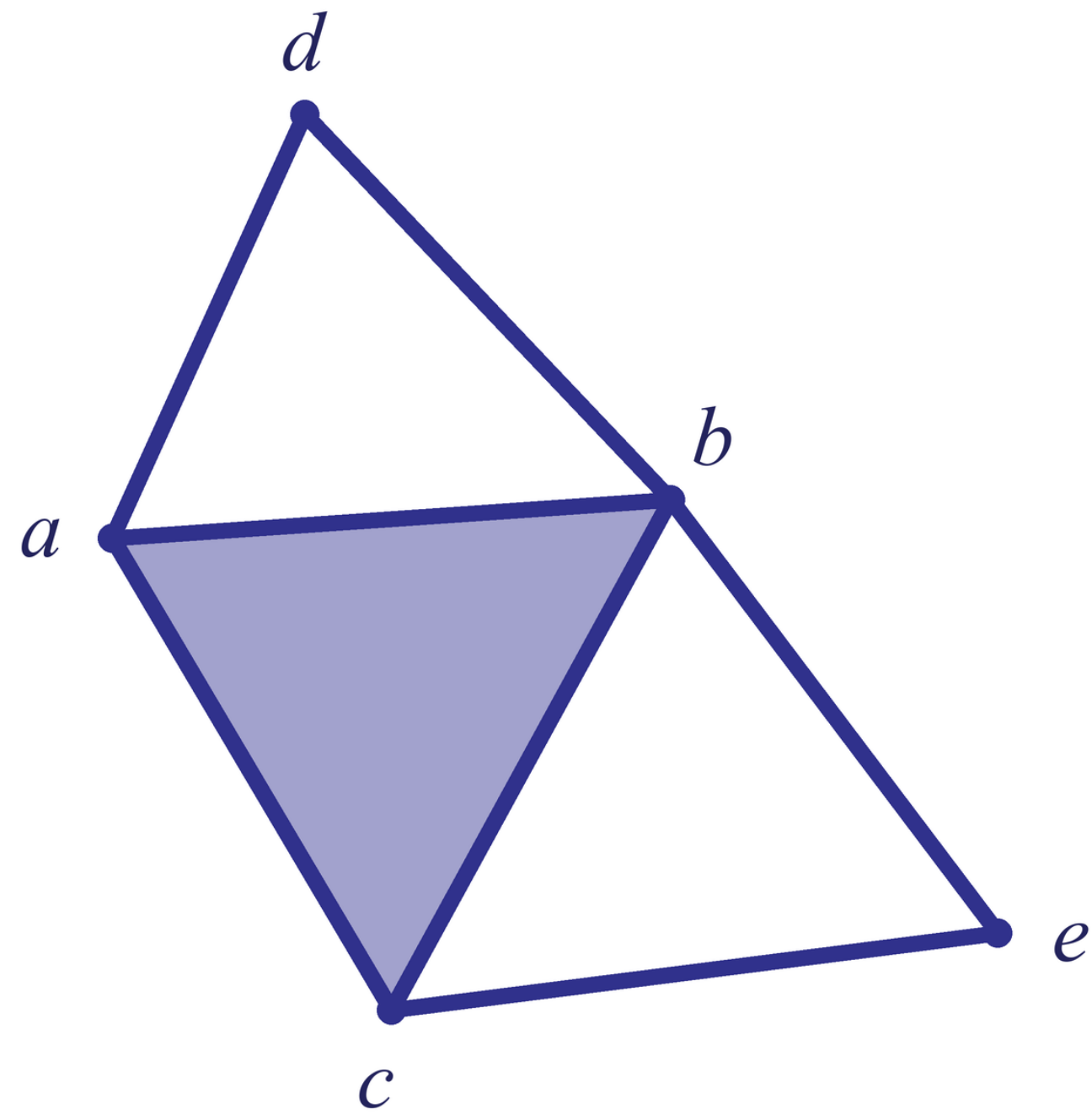
The collection of p -boundaries forms a subspace $B_p(K) \subseteq C_p(K)$

Example: Cycles and Boundaries

What are the generators of $B_1(K)$? Of $Z_1(K)$?



Example: Cycles and Boundaries



What are the generators of $B_1(K)$? Of $Z_1(K)$?

Homology for Real Now

Equivalence Classes

An equivalence relation $\sim$ on a set S satisfies:

- Reflexivity: $x \sim x$
- Symmetry: if $x \sim y$, then $y \sim x$
- Transitivity: if $x \sim y$ and $y \sim z$, then $x \sim z$

The equivalence class of x consists of all elements equivalent to x :

$$[x] = \{y \in S \mid y \sim x\}$$

Ex. On $\mathbb{Z}$, define $m \sim n$ when $m - n$ is even.

Quotient Space

Let V be a vector space over a field k . Let $W \subset V$ be a subspace.

Define $\sim$ on V by $x \sim y$ iff $x - y \in W$. The equivalence class of x is denoted

$$[x] = x + W = \{x + w : w \in W\}$$

The quotient space V/W is then defined as $\{[x] \mid x \in V\}$. This is also a vector space with:

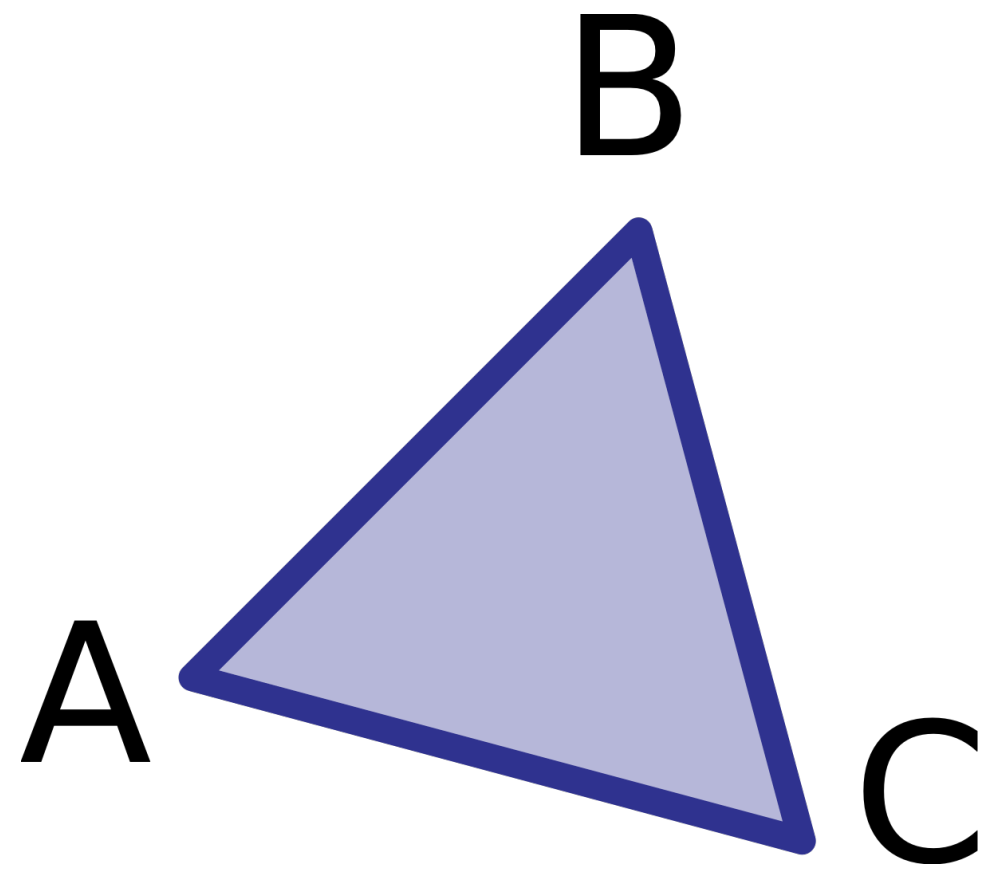
- Scalar multiplication:
- Addition:

Homology!!

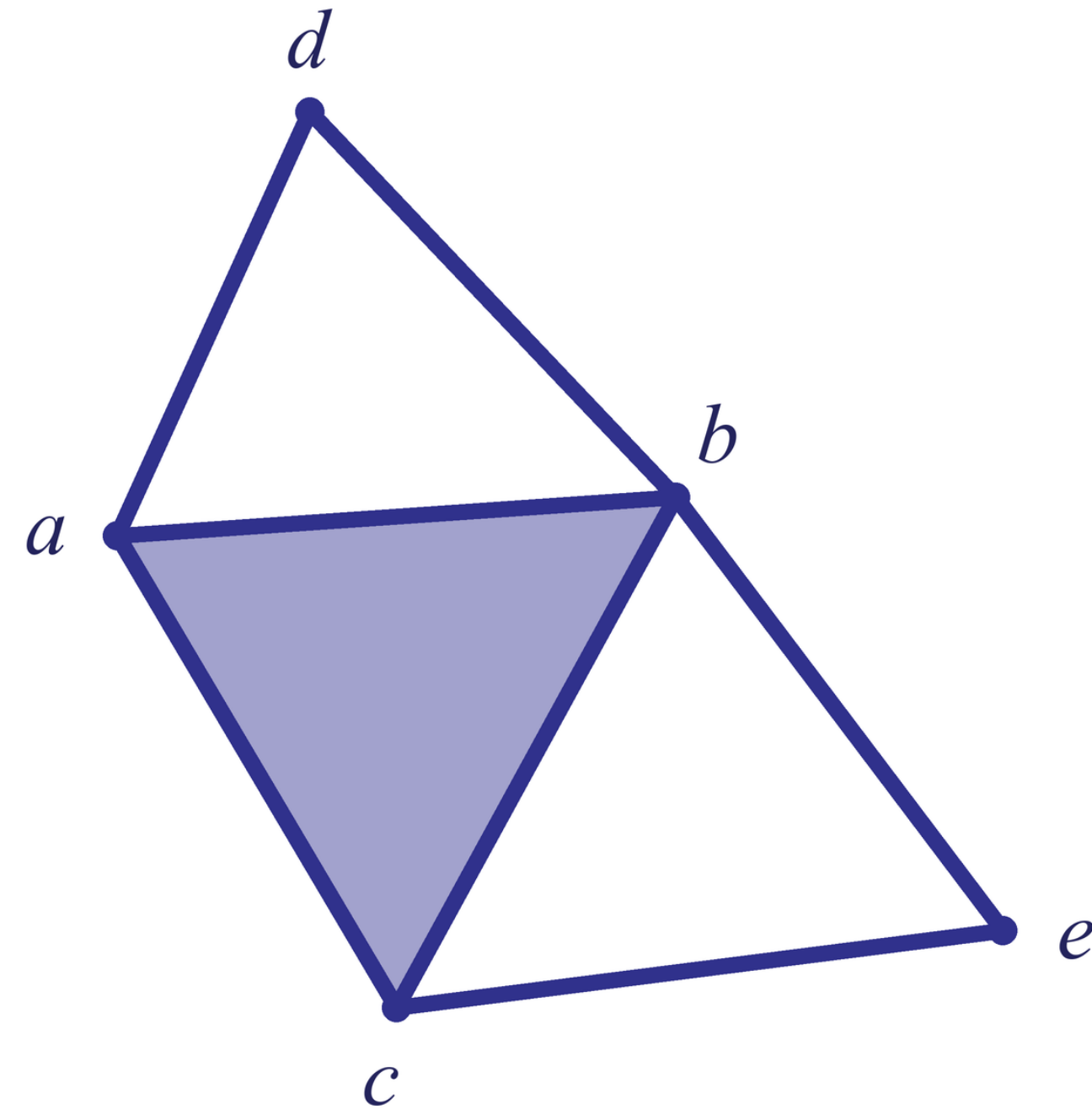
The *p*th homology group is the quotient space

$$H_p(K) := Z_p(K)/B_p(K)$$

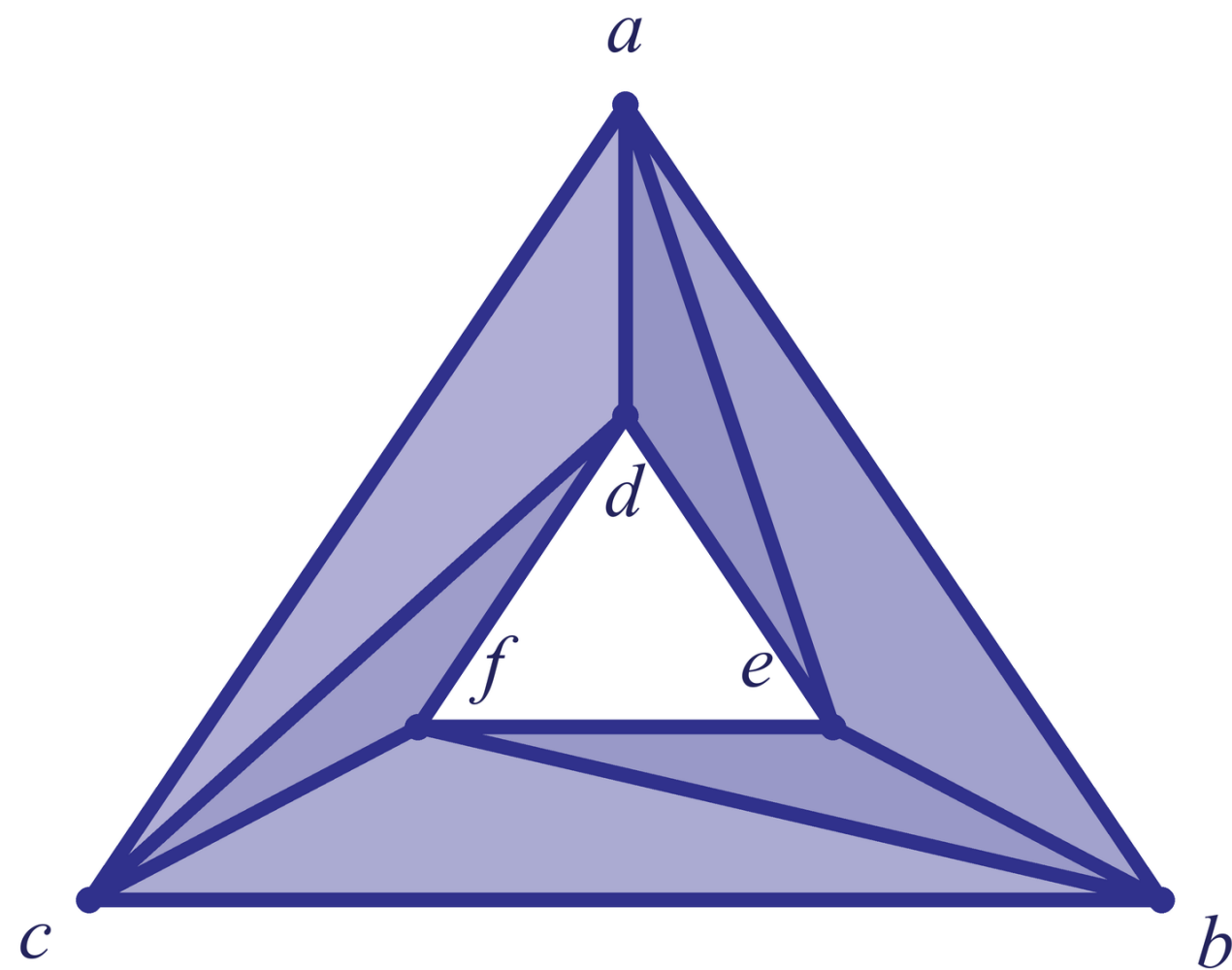
Example: What is $H_1(K)$?



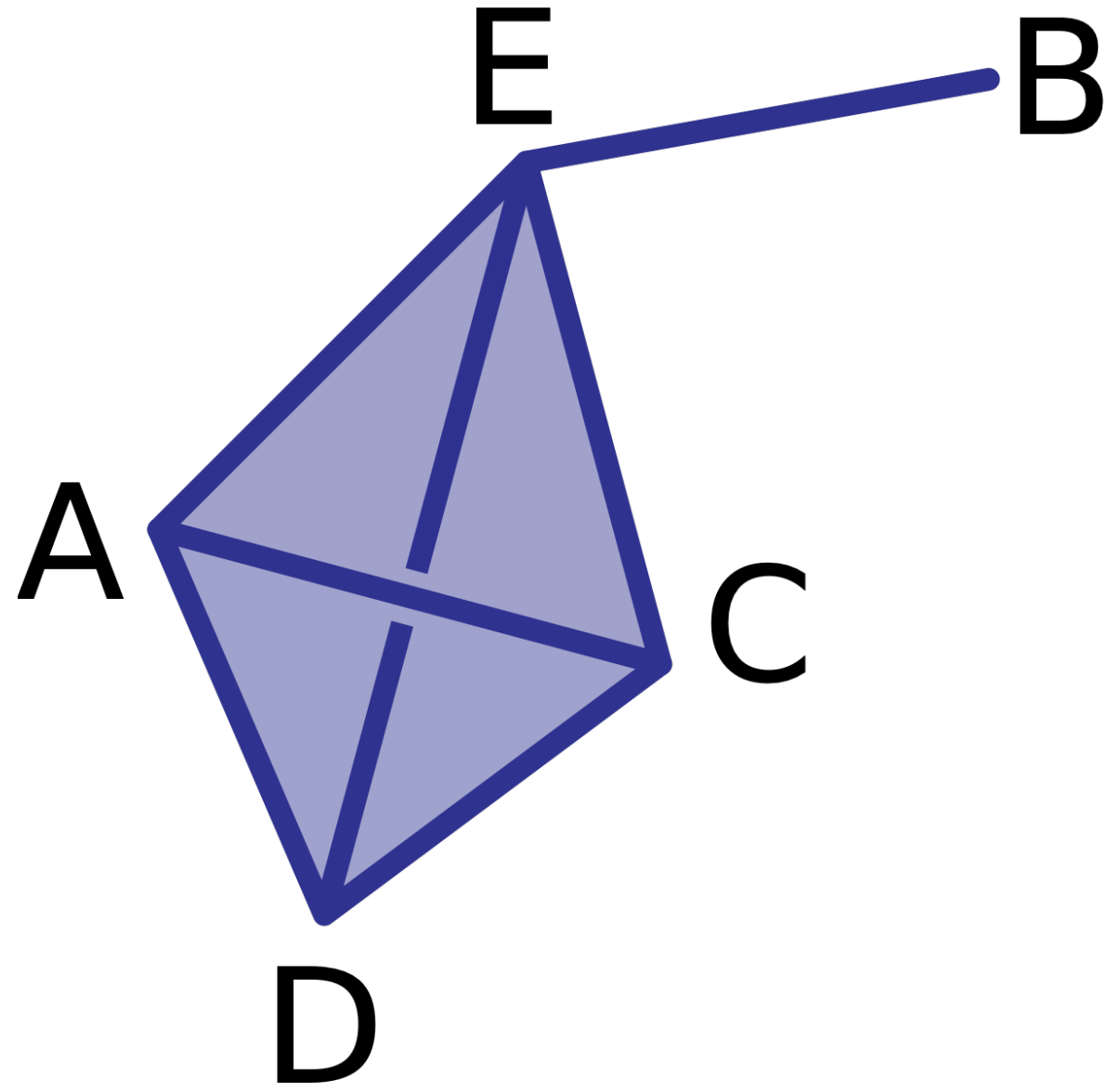
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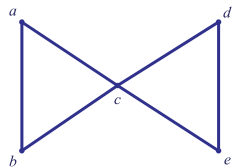
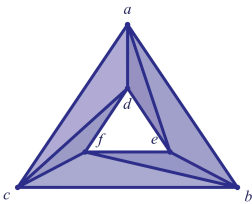
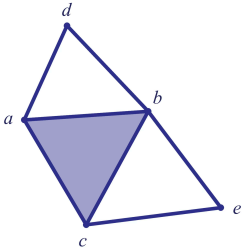
Example: What is $H_2(K)$?



Betti Numbers

$\beta_k = \dim H_k = \dim Z_k - \dim B_k$

- β_0 : number of connected components
- β_1 : number of independent loop classes
- β_2 : number of independent 2-dimensional homology classes

	β_0	β_1	β_2
			
			
			

Take away

- We want to compute the “shape”: we turn “shape” into linear algebra; homology is the machinery
- p -simplex: basis vector
- p -chains: vectors
- Boundary map: matrix
- Cycles: kernel of a matrix
- Boundaries: image of a matrix
- Homology: cycles modulo trivial cycles

Homework from last time

- Let K be the simplicial complex of a tetrahedron. Write a basis for the chain groups C_1 and C_2 ; the boundary groups B_1 and B_2 ; and the cycle groups Z_1 and Z_2 . Write the boundary matrix representing the boundary operator ∂_2 , with rows and columns representing bases of C_1 and C_2 , respectively.
- Consider a complex $K = \{A, B, C, D, E, AB, BC, AC, CD, AD, CE, DE, ABC, ACD\}$. Write a basis for the chain groups C_1 and C_2 ; the boundary groups B_1 and B_2 ; and the cycle groups Z_1 and Z_2 ; and write the matrix ∂_2 . Finally, identify a 1-cycle that is not a 1-boundary.